Application of Derivative: Questions and Full Solutions

Question 01

English

Let P be a point on the parabola $x^2 = 4y$ such that exactly two distinct lines passing through P are tangent to the curve $y = x^3$. Then the y-intercept of the tangent to the parabola $x^2 = 4y$ at P is equal to

(A) $\frac{1}{64}$
(B) $-\frac{1}{32}$
(C) $-\frac{1}{64}$
(D) $-\frac{1}{16}$

Only one option is correct.

हिन्दी

माना कि P परवलय $x^2 = 4y$ पर एक ऐसा बिन्दु है कि P से होकर जाने वाली ठीक दो भिन्न रेखाएँ वक्र $y = x^3$ की स्पर्श रेखाएँ हैं। तब परवलय $x^2 = 4y$ के बिन्दु P पर खींची गई स्पर्श रेखा का y-अन्तःखण्ड बराबर है

(A) $\frac{1}{64}$
(B) $-\frac{1}{32}$
(C) $-\frac{1}{64}$
(D) $-\frac{1}{16}$

केवल एक विकल्प सही है।

Solution

Any point of $x^2 = 4y$ can be written as $P(2s, s^2)$. Since $y' = \frac{x}{2}$, the tangent at P has slope s, and its equation is $y - s^2 = s(x - 2s)$, that is $y = sx - s^2$ (slope form of the tangent of $x^2 = 4y$).

Write $P = (h, k)$ with $h = 2s$, $k = s^2$. The tangent to $y = x^3$ at (a, a^3) is $y - a^3 = 3a^2(x - a)$. It passes through (h, k) iff $k - a^3 = 3a^2(h - a)$, i.e. $g(a) = 2a^3 - 3ha^2 + k = 0$.

Different real roots a give different tangent lines (the slopes $3a^2$ with the intercepts $-2a^3$ differ), so the number of tangents through P equals the number of distinct real roots of g.

$g'(a) = 6a^2 - 6ha = 6a(a - h)$, so for $h \neq 0$ the critical points are $a = 0$ and $a = h$, with $g(0) = k$ and $g(h) = k - h^3$.

A cubic has three distinct real roots when $g(0)g(h) < 0$, one real root when $g(0)g(h) > 0$, and exactly two distinct roots (one of them double) when exactly one of $g(0)$, $g(h)$ is zero.

Case $k = 0$: $s^2 = 0 \Rightarrow s = 0$, so $P = (0, 0)$; then $h = 0$ and $g(a) = 2a^3$ has the single root $a = 0$. Only one tangent, so this case is rejected.

Case $k = h^3$: $s^2 = 8s^3 \Rightarrow s^2(8s - 1) = 0 \Rightarrow s = \frac{1}{8}$ (as $s \neq 0$). Then $P = \left(\frac{1}{4}, \frac{1}{64}\right)$, which lies on $y = x^3$.

Check: $g(a) = 2a^3 - \frac{3}{4}a^2 + \frac{1}{64} = 2\left(a - \frac{1}{4}\right)^2\left(a + \frac{1}{8}\right)$, so the two tangents are those at $a = \frac{1}{4}$ and $a = -\frac{1}{8}$.

For every other s, $g(0) = s^2 > 0$ and $g(h) = s^2(1 - 8s)$ is non-zero, so the count is 1 or 3; hence P is unique.

The tangent to the parabola at P is $y = \frac{1}{8}x - \frac{1}{64}$; its y-intercept is $-\frac{1}{64}$.

Correct Option: (C)

Question 02

English

A rectangle R with sides parallel to the coordinate axes is inscribed in the circle $x^2 + y^2 = 25$. Its vertex $V(x, y)$ in the first quadrant moves on the circle so that $\frac{dx}{dt} = -1$ unit per second. Using differentials, the approximate area of R (in square units) 0.1 second after the instant at which V is at $(3, 4)$ is

(A) 47.3
(B) 45.5
(C) 48.7
(D) 47.825

Only one option is correct.

हिन्दी

निर्देशांक अक्षों के समान्तर भुजाओं वाला एक आयत R वृत्त $x^2 + y^2 = 25$ के अन्तर्गत है। प्रथम चतुर्थांश में इसका शीर्ष $V(x, y)$ वृत्त पर इस प्रकार गति करता है कि $\frac{dx}{dt} = -1$ इकाई प्रति सेकण्ड। अवकलों का उपयोग करके, जिस क्षण V बिन्दु $(3, 4)$ पर है उससे 0.1 सेकण्ड बाद R का सन्निकट क्षेत्रफल (वर्ग इकाई में) है

(A) 47.3
(B) 45.5
(C) 48.7
(D) 47.825

केवल एक विकल्प सही है।

Solution

The rectangle has vertices $(\pm x, \pm y)$, so its sides are $2x$ and $2y$ and its area is $A = 4xy$. At $V(3, 4)$, $A = 4 \cdot 3 \cdot 4 = 48$.

Since V stays on the circle, $x^2 + y^2 = 25$ for all t. Differentiating with respect to t: $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$.

At $(3, 4)$ with $\frac{dx}{dt} = -1$: $3(-1) + 4\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = \frac{3}{4}$ (the vertex rises as it moves left, as it must on the circle).

By the product rule, $\frac{dA}{dt} = 4\left(\frac{dx}{dt}\,y + x\,\frac{dy}{dt}\right) = 4\left((-1)(4) + 3 \cdot \frac{3}{4}\right) = 4\left(-4 + \frac{9}{4}\right) = -7$ square units per second.

Treating A as a function of t, the differential gives $\Delta A \approx dA = \frac{dA}{dt}\,\Delta t$ with $\Delta t = 0.1$: $\Delta A \approx -0.7$.

Hence the approximate area after 0.1 second is $48 - 0.7 = 47.3$ square units.

(For comparison, the exact area is about 47.25, so the approximation is sound.)

Correct Option: (A)

Question 03

English

Let $A(0,1)$ and $B(0,3)$ be two points and let $P(x,0)$, $x>2$, be a point on the x-axis. The angle $\theta=\angle APB$ is measured as $\tan^{-1}\left(\frac{1}{2}\right)$ with a possible error of 0.002 radian. Using differentials, the approximate magnitude of the resulting error in the value of x computed from θ is

(A) 0.01
(B) 0.03
(C) 0.024
(D) 0.006

Only one option is correct.

हिन्दी

माना कि $A(0,1)$ तथा $B(0,3)$ दो बिन्दु हैं और $P(x,0)$, $x>2$, x-अक्ष पर एक बिन्दु है। कोण $\theta=\angle APB$ को $\tan^{-1}\left(\frac{1}{2}\right)$ मापा गया है, जिसमें 0.002 रेडियन की सम्भावित त्रुटि है। अवकलों का उपयोग करके, θ से परिकलित x के मान में उत्पन्न त्रुटि का सन्निकट परिमाण है

(A) 0.01
(B) 0.03
(C) 0.024
(D) 0.006

केवल एक विकल्प सही है।

Solution

With O the origin, $\angle OPB=\tan^{-1}\frac{3}{x}$ and $\angle OPA=\tan^{-1}\frac{1}{x}$ (right triangles OPB, OPA), and $\theta=\angle OPB-\angle OPA$.

By $\tan(\alpha-\beta)=\dfrac{\tan\alpha-\tan\beta}{1+\tan\alpha\tan\beta}$: $\tan\theta=\dfrac{\frac{3}{x}-\frac{1}{x}}{1+\frac{3}{x^2}}=\dfrac{2x}{x^2+3}$.

Measured value: $\dfrac{2x}{x^2+3}=\dfrac{1}{2}\Rightarrow x^2-4x+3=0\Rightarrow x=1$ or $x=3$. Both positions see the same angle (they are on opposite sides of $x=\sqrt{3}$, where θ is largest); the condition $x>2$ selects $x=3$.

Differentiate $\tan\theta=\dfrac{2x}{x^2+3}$: $\sec^2\theta\,\dfrac{d\theta}{dx}=\dfrac{2(x^2+3)-2x\cdot 2x}{(x^2+3)^2}=\dfrac{2(3-x^2)}{(x^2+3)^2}$.

At $x=3$: right side $=\dfrac{2(3-9)}{144}=-\dfrac{1}{12}$, and $\sec^2\theta=1+\dfrac{1}{4}=\dfrac{5}{4}$. So $\dfrac{d\theta}{dx}=-\dfrac{1}{12}\cdot\dfrac{4}{5}=-\dfrac{1}{15}$, i.e. $dx=-15\,d\theta$.

With $|d\theta|=0.002$: $|dx|\approx 15\times 0.002=0.03$.

Check of the trap: at $x=1$, $\dfrac{d\theta}{dx}=\dfrac{2\cdot 2}{16}\cdot\dfrac{4}{5}=\dfrac{1}{5}$ and $|dx|=0.01$, which is option (A); it is excluded by $x>2$.

Correct Option: (B)

Question 04

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x + \log_e\left(x^2 - 2x + 2\right)$. Then which of the following is TRUE?

(A) $e < \left(\frac{18}{13}\right)^3$

(B) $e < \left(\frac{25}{18}\right)^3$

(C) $\sqrt{e} < \frac{8}{5}$

(D) $e > \frac{169}{64}$

Only one option is correct.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x + \log_e\left(x^2 - 2x + 2\right)$ द्वारा परिभाषित है। तब निम्न में से कौन-सा सत्य है?

(A) $e < \left(\frac{18}{13}\right)^3$

(B) $e < \left(\frac{25}{18}\right)^3$

(C) $\sqrt{e} < \frac{8}{5}$

(D) $e > \frac{169}{64}$

केवल एक विकल्प सही है।

Solution

Domain: $x^2 - 2x + 2 = (x-1)^2 + 1 \geq 1 > 0$ for all real x, so f is defined and differentiable on $\mathbb{R}$.

$$f'(x) = 1 + \frac{2x-2}{x^2-2x+2} = \frac{x^2-2x+2+2x-2}{x^2-2x+2} = \frac{x^2}{x^2-2x+2}.$$

So $f'(x) \geq 0$ for all x, with equality only at the single point $x = 0$. Hence f is strictly increasing on $\mathbb{R}$ (always increasing), and $a < b \Rightarrow f(a) < f(b)$. Also $f(0) = \log_e 2$.

(A) $f\left(\frac{1}{3}\right) = \frac{1}{3} + \log_e \frac{13}{9}$. Since $\frac{1}{3} > 0$, $f\left(\frac{1}{3}\right) > \log_e 2$, i.e. $\frac{1}{3} > \log_e \frac{18}{13}$, so $e^{1/3} > \frac{18}{13}$ and $e > \left(\frac{18}{13}\right)^3$. Statement (A) says the opposite, so (A) is wrong.

(B) $f\left(-\frac{1}{3}\right) = -\frac{1}{3} + \log_e \frac{25}{9} < \log_e 2$, so $\log_e \frac{25}{18} < \frac{1}{3}$, i.e. $e > \left(\frac{25}{18}\right)^3$. So (B) is wrong.

(C) $f\left(\frac{1}{2}\right) = \frac{1}{2} + \log_e \frac{5}{4} > \log_e 2$, so $\frac{1}{2} > \log_e \frac{8}{5}$, i.e. $\sqrt{e} > \frac{8}{5}$. So (C) is wrong.

(D) $f\left(-\frac{1}{2}\right) = -\frac{1}{2} + \log_e \frac{13}{4} < \log_e 2$, so $\log_e \frac{13}{8} < \frac{1}{2}$, i.e. $\sqrt{e} > \frac{13}{8}$ and $e > \frac{169}{64}$. So (D) is correct.

Each wrong option is exactly the reversed inequality obtained by treating f as decreasing.

Correct Option: (D)

Question 05

English

Let P be a point in the first quadrant on the curve $y = x^2$ such that the normal to the curve at P passes through the point $\left(0, \frac{9}{2}\right)$. The tangent and the normal to the curve at P meet the x-axis at T and G, respectively. Then the area of the triangle PTG is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि वक्र $y = x^2$ पर प्रथम चतुर्थांश में P एक ऐसा बिन्दु है कि P पर वक्र का अभिलम्ब बिन्दु $\left(0, \frac{9}{2}\right)$ से होकर जाता है। P पर वक्र की स्पर्श रेखा तथा अभिलम्ब x-अक्ष को क्रमशः T तथा G पर मिलते हैं। तब त्रिभुज PTG का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $y = x^2$, $y' = 2x$. At $P(a, a^2)$, $a > 0$, the slope of the tangent is $2a$ and the normal is $y - a^2 = -\dfrac{1}{2a}(x - a)$, i.e. $(x - a) + 2a(y - a^2) = 0$.

It passes through $\left(0, \frac{9}{2}\right)$: $-a + 2a\left(\frac{9}{2} - a^2\right) = 0 \Rightarrow a\left(8 - 2a^2\right) = 0$. The root $a = 0$ is not in the first quadrant, so $a^2 = 4$, $a = 2$, $P(2, 4)$ and the slope is $m = 4$.

Let $M(2, 0)$ be the foot of the ordinate of P. The subtangent is $TM = \left|\dfrac{y}{m}\right| = \dfrac{4}{4} = 1$, so $T = (1, 0)$ (check: tangent $y = 4x - 4$ meets $y = 0$ at $x = 1$).

The subnormal is $MG = |y\, m| = 4 \cdot 4 = 16$, so $G = (18, 0)$ (check: normal $y - 4 = -\frac{1}{4}(x - 2)$ meets $y = 0$ at $x = 18$).

T and G lie on opposite sides of M, so $TG = TM + MG = 17$, and the height of the triangle from P is the ordinate 4.

Area of $\triangle PTG = \dfrac{1}{2} \cdot 17 \cdot 4 = 34$.

Final Answer: 34

Question 06

English

Let c be a real number and let $C_1 : y = x^2$ and $C_2 : y = -x^2 + 5x + c$. If each of the tangents drawn from the point $\left(\frac{5}{4}, 1\right)$ to C_1 also touches C_2, then the value of c is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि c एक वास्तविक संख्या है तथा $C_1 : y = x^2$ और $C_2 : y = -x^2 + 5x + c$ हैं। यदि बिन्दु $\left(\frac{5}{4}, 1\right)$ से C_1 पर खींची गई प्रत्येक स्पर्श रेखा C_2 को भी स्पर्श करती है, तो c का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The tangent to $C_1 : y = x^2$ at (a, a^2) is $y = 2ax - a^2$.

It passes through the external point $\left(\frac{5}{4}, 1\right)$ iff $1 = \frac{5}{2}a - a^2$, i.e. $2a^2 - 5a + 2 = 0$, so $a = 2$ or $a = \frac{1}{2}$. ($1 < \left(\frac{5}{4}\right)^2$, so the point is outside C_1 and there are two tangents.)

The two tangents are $L_1 : y = 4x - 4$ and $L_2 : y = x - \frac{1}{4}$.

L_1 touches C_2 iff $-x^2 + 5x + c = 4x - 4$, i.e. $x^2 - x - (c + 4) = 0$, has a repeated root: $1 + 4(c + 4) = 0 \Rightarrow c = -\frac{17}{4}$.

L_2 touches C_2 iff $x^2 - 4x - \left(c + \frac{1}{4}\right) = 0$ has a repeated root: $16 + 4\left(c + \frac{1}{4}\right) = 0 \Rightarrow c = -\frac{17}{4}$.

Both conditions give the same value, so the requirement is consistent and $c = -\frac{17}{4} = -4.25$.

(Reason for the consistency: with $c = -\frac{17}{4}$, $C_2 : y = 2 - \left(\frac{5}{2} - x\right)^2$ is the image of C_1 under the point reflection in $\left(\frac{5}{4}, 1\right)$, which maps every line through that point to itself.)

Final Answer: -4.25

Question 07

English

The parabolas $y^2 = 8x$ and $x^2 = -8y$ have exactly one common normal. If this common normal meets the y-axis at the point $(0, \beta)$, then the value of β is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

परवलयों $y^2 = 8x$ तथा $x^2 = -8y$ का ठीक एक उभयनिष्ठ अभिलम्ब है। यदि यह उभयनिष्ठ अभिलम्ब y-अक्ष को बिन्दु $(0, \beta)$ पर मिलता है, तो β का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $y^2 = 4ax$ with $a = 2$, the normal with slope m is $y = mx - 2am - am^3 = mx - 4m - 2m^3$; it is normal at $(2m^2, -4m)$. The case $m = 0$ is the axis $y = 0$.

For $x^2 = -8y$, i.e. $y = -\dfrac{x^2}{8}$, take the point $(4s, -2s^2)$. Then $y' = -\dfrac{x}{4} = -s$, so for $s \neq 0$ the normal has slope $\dfrac{1}{s}$: $y + 2s^2 = \dfrac{1}{s}(x - 4s)$, i.e. $y = \dfrac{x}{s} - 4 - 2s^2$. At $s = 0$ the normal is $x = 0$.

The line $y = 0$ is not of the second form and $x = 0$ is not of the first, so a common normal needs $m = \dfrac{1}{s}$ and equal intercepts: $-\dfrac{4}{s} - \dfrac{2}{s^3} = -4 - 2s^2$.

Multiplying by $-\dfrac{s^3}{2}$: $2s^2 + 1 = 2s^3 + s^5$, so $s^5 + 2s^3 - 2s^2 - 1 = 0$. Since $s = 1$ is a root, $s^5 + 2s^3 - 2s^2 - 1 = (s - 1)(s^4 + s^3 + 3s^2 + s + 1)$.

$s^4 + s^3 + 3s^2 + s + 1 = \left(s^2 + \dfrac{s}{2}\right)^2 + \dfrac{11}{4}s^2 + s + 1$, and $\dfrac{11}{4}s^2 + s + 1$ has discriminant $1 - 11 < 0$, so the quartic has no real root. Hence $s = 1$ and $m = 1$ only.

The common normal is $y = x - 6$ (normal to $y^2 = 8x$ at $(2, -4)$ and to $x^2 = -8y$ at $(4, -2)$). It is symmetric about the line $y = -x$, which maps one parabola to the other, as expected.

It meets the y-axis at $(0, -6)$, so $\beta = -6$.

Final Answer: -6

Question 08

English

Consider the curve C given by $x = t^2$, $y = t^3 - 3t$, $t \in \mathbb{R}$. The tangent to C at the point with parameter $t = 1$ meets C again at the point Q. If the normal to C at Q meets the x-axis at $(\alpha, 0)$, then the value of α is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वक्र C पर विचार कीजिए जो $x = t^2$, $y = t^3 - 3t$, $t \in \mathbb{R}$ द्वारा दिया गया है। प्राचल $t = 1$ वाले बिन्दु पर C की स्पर्श रेखा C को पुनः बिन्दु Q पर मिलती है। यदि Q पर C का अभिलम्ब x-अक्ष को $(\alpha, 0)$ पर मिलता है, तो α का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 3t^2 - 3$, so for $t \neq 0$, $\frac{dy}{dx} = \frac{3(t^2-1)}{2t}$.

At $t = 1$: point $(1, -2)$, slope 0, so the tangent is the horizontal line $y = -2$.

Points of C on $y = -2$: $t^3 - 3t + 2 = 0 \Rightarrow (t-1)^2(t+2) = 0$. The double root $t = 1$ is the point of contact, so the tangent meets C again only at $t = -2$, i.e. $Q = (4, -2)$.

At $t = -2$: $\frac{dy}{dx} = \frac{3(4-1)}{-4} = -\frac{9}{4}$, so the slope of the normal at Q is $\frac{4}{9}$.

Normal at Q: $y + 2 = \frac{4}{9}(x - 4)$. Putting $y = 0$: $2 = \frac{4}{9}(x-4) \Rightarrow x = 4 + \frac{9}{2} = \frac{17}{2}$.

Hence $\alpha = \frac{17}{2} = 8.5$.

Final Answer: 8.5

Question 09

English

Let d denote the minimum distance between the curves $y = x^2$ and $y = -\left(x - \frac{3}{2}\right)^2$. Then the value of d^2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि d वक्रों $y = x^2$ तथा $y = -\left(x - \frac{3}{2}\right)^2$ के बीच की न्यूनतम दूरी को दर्शाता है। तब d^2 का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The first curve lies on or above the x-axis and the second on or below it, and they cannot both be 0 at the same x, so the curves are disjoint and the minimum distance is positive.

Let $P(a, a^2)$ and $Q\left(b, -\left(b - \frac{3}{2}\right)^2\right)$ be the nearest points. Then PQ is normal to both curves, so the tangent slopes are equal: $2a = -2\left(b - \frac{3}{2}\right)$, i.e. $b = \frac{3}{2} - a$ and $Q = \left(\frac{3}{2} - a, -a^2\right)$.

$\overrightarrow{PQ} = \left(\frac{3}{2} - 2a, -2a^2\right)$ must be parallel to the normal direction $(-2a, 1)$ of $y = x^2$ at P: $\left(\frac{3}{2} - 2a\right) \cdot 1 - (-2a^2)(-2a) = 0$, i.e. $4a^3 + 2a - \frac{3}{2} = 0$, or $8a^3 + 4a - 3 = 0$.

$8a^3 + 4a - 3 = (2a - 1)(4a^2 + 2a + 3)$ and $4a^2 + 2a + 3$ has negative discriminant, so $a = \frac{1}{2}$: $P\left(\frac{1}{2}, \frac{1}{4}\right)$, $Q\left(1, -\frac{1}{4}\right)$.

Why this is the minimum: the tangent to $y = x^2$ at P is $y = x - \frac{1}{4}$ and the tangent to the second curve at Q is $y = x - \frac{5}{4}$. The convex curve $y = x^2$ lies above its tangent and the concave curve lies below its tangent, so every segment joining the curves crosses the strip between these parallel lines, whose width is $\frac{1}{\sqrt{2}}$. Hence the distance is at least $\frac{1}{\sqrt{2}}$, and $PQ = \frac{1}{\sqrt{2}}$ attains it.

So $d = \frac{1}{\sqrt{2}}$ and $d^2 = \frac{1}{2} = 0.5$.

Final Answer: 0.5

Question 10

English

Let $\alpha > 0$ and let S be the circle $(x-\alpha)^2 + y^2 = 4$. If the minimum distance between S and the parabola $y^2 = 12x$ is 40, then the value of α is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $\alpha > 0$ तथा S वृत्त $(x-\alpha)^2 + y^2 = 4$ है। यदि S तथा परवलय $y^2 = 12x$ के बीच की न्यूनतम दूरी 40 है, तो α का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For a circle with centre C and radius r lying in the region $y^2 < 12x$ and not meeting the parabola, the minimum distance to the parabola is (least distance of C from the parabola) $-r$, attained along the normal to the parabola through C.

Least distance of $C(\alpha, 0)$ from $y^2 = 12x$: for a point (x, y) of the parabola, $D^2 = (x-\alpha)^2 + y^2 = (x-\alpha)^2 + 12x$, $x \geq 0$.

$\dfrac{d(D^2)}{dx} = 2(x-\alpha) + 12$, which vanishes at $x = \alpha - 6$; $\dfrac{d^2(D^2)}{dx^2} = 2 > 0$, so this is the minimum when $\alpha - 6 \geq 0$.

If $0 < \alpha \leq 6$, D^2 is increasing on $x \geq 0$, the least distance is $D(0) = \alpha \leq 6$, and the circle-parabola distance is at most 4, not 40. So $\alpha > 6$.

For $\alpha > 6$: least $D^2 = (-6)^2 + 12(\alpha - 6) = 12\alpha - 36$, and the circle lies inside the parabola's region because $\sqrt{12\alpha - 36} > 2$.

Given distance: $\sqrt{12\alpha - 36} - 2 = 40 \Rightarrow \sqrt{12\alpha - 36} = 42 \Rightarrow 12\alpha = 1764 + 36 = 1800 \Rightarrow \alpha = 150$.

(The condition $\alpha > 0$ excludes the second configuration $\alpha = -42$, where the centre is outside the parabola and the nearest point is the vertex.)

Final Answer: 150

Question 11

English

Let p and q be real numbers such that the parabolas $y = x^2$ and $y = 2x^2 + px + q$ intersect orthogonally at the point whose abscissa is $\frac{\sqrt{2}}{4}$. Then the product of the slopes of the common tangents of these two parabolas is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि p तथा q ऐसी वास्तविक संख्याएँ हैं कि परवलय $y = x^2$ तथा $y = 2x^2 + px + q$ उस बिन्दु पर लम्बवत् प्रतिच्छेद करते हैं जिसका भुज $\frac{\sqrt{2}}{4}$ है। तब इन दोनों परवलयों की उभयनिष्ठ स्पर्श रेखाओं की प्रवणताओं का गुणनफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $x_0 = \frac{\sqrt{2}}{4}$. The slope of $y = x^2$ at x_0 is $2x_0 = \frac{\sqrt{2}}{2}$, and the slope of $y = 2x^2 + px + q$ is $4x_0 + p = \sqrt{2} + p$.

The curves cut orthogonally, so the product of the slopes is -1: $\frac{\sqrt{2}}{2}(\sqrt{2} + p) = -1 \Rightarrow 1 + \frac{\sqrt{2}}{2}p = -1 \Rightarrow p = -2\sqrt{2}$.

x_0 is a common point: $x_0^2 = 2x_0^2 + px_0 + q \Rightarrow q = -x_0^2 - px_0 = -\frac{1}{8} + 2\sqrt{2} \cdot \frac{\sqrt{2}}{4} = \frac{7}{8}$.

Every tangent of $y = x^2$ is $y = 2ax - a^2$ for some a. It touches $y = 2x^2 + px + q$ iff $2x^2 + (p - 2a)x + (q + a^2) = 0$ has a repeated root: $(p - 2a)^2 - 8(q + a^2) = 0$, i.e. $4a^2 + 4pa - p^2 + 8q = 0$.

With $p = -2\sqrt{2}$, $q = \frac{7}{8}$: $4a^2 - 8\sqrt{2}a - 8 + 7 = 0$, i.e. $4a^2 - 8\sqrt{2}a - 1 = 0$, whose discriminant $128 + 16 > 0$ gives two common tangents.

Their slopes are $2a_1$, $2a_2$ with $a_1 a_2 = -\frac{1}{4}$, so the product of the slopes is $4a_1 a_2 = -1$ (the common tangents are perpendicular; explicitly the slopes are $2\sqrt{2} \pm 3$).

Final Answer: -1

Question 12

English

Let $f : (0, \infty) \to (0, \infty)$ be a differentiable increasing function with $f(2) = 1$, and let C be the curve $y = f(x)$. For every point $P(x, y)$ on C, the length of the subtangent at P is equal to the radius of the circle that passes through the origin and P and has its centre on the x-axis. If the point $(\alpha, 6)$ lies on C, then the value of α^2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : (0, \infty) \to (0, \infty)$ एक अवकलनीय वर्धमान फलन है जिसके लिए $f(2) = 1$ है, तथा C वक्र $y = f(x)$ है। C के प्रत्येक बिन्दु $P(x, y)$ के लिए, P पर अधःस्पर्शी (subtangent) की लम्बाई उस वृत्त की त्रिज्या के बराबर है जो मूलबिन्दु तथा P से होकर जाता है और जिसका केन्द्र x-अक्ष पर है। यदि बिन्दु $(\alpha, 6)$ वक्र C पर स्थित है, तो α^2 का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let the circle through $O(0, 0)$ and $P(x, y)$ have centre $(h, 0)$. Equal radii: $h^2 = (x - h)^2 + y^2 \Rightarrow 0 = x^2 - 2hx + y^2 \Rightarrow h = \dfrac{x^2 + y^2}{2x}$. Since $x > 0$, the radius is $\dfrac{x^2 + y^2}{2x}$.

The subtangent at P has length $\left|\dfrac{y}{y'}\right|$. Here $y = f(x) > 0$ and f is increasing, so $y' > 0$ (where y' would vanish the subtangent is undefined, which the condition excludes) and the length is $\dfrac{y}{y'}$.

Condition: $\dfrac{y}{y'} = \dfrac{x^2 + y^2}{2x}$, i.e. $\dfrac{dy}{dx} = \dfrac{2xy}{x^2 + y^2}$, a homogeneous differential equation.

Put $y = vx$: $v + x\dfrac{dv}{dx} = \dfrac{2v}{1 + v^2}$, so $x\dfrac{dv}{dx} = \dfrac{2v - v - v^3}{1 + v^2} = \dfrac{v(1 - v^2)}{1 + v^2}$.

Separate: $\dfrac{1 + v^2}{v(1 - v^2)}dv = \dfrac{dx}{x}$. Partial fractions: $\dfrac{1 + v^2}{v(1 - v)(1 + v)} = \dfrac{1}{v} + \dfrac{1}{1 - v} - \dfrac{1}{1 + v}$.

Integrating: $\ln|v| - \ln|1 - v| - \ln|1 + v| = \ln x + c$, so $\dfrac{v}{1 - v^2} = kx$ for a constant k, i.e. $\dfrac{xy}{x^2 - y^2} = kx$, which gives $x^2 - y^2 = \dfrac{y}{k}$.

At $(2, 1)$: $4 - 1 = \dfrac{1}{k} \Rightarrow \dfrac{1}{k} = 3$, so $C : \ x^2 - y^2 = 3y$, i.e. $y = \dfrac{-3 + \sqrt{9 + 4x^2}}{2}$, which is positive and increasing on $(0, \infty)$, as required.

Check: from $2x - 2yy' = 3y'$, $y' = \dfrac{2x}{2y + 3}$ and $\dfrac{y}{y'} = \dfrac{2y^2 + 3y}{2x} = \dfrac{y^2 + (y^2 + 3y)}{2x} = \dfrac{x^2 + y^2}{2x}$.

For $(\alpha, 6)$ on C: $\alpha^2 = 36 + 18 = 54$.

Final Answer: 54

Question 13

English

A sector of a circle has a constant area of 25 cm^2, while its radius r and its central angle vary with time. At the instant when $r = 4$ cm, the central angle is decreasing at the rate of $\frac{45}{\pi}$ degrees per second. Then the rate of change of the perimeter of the sector (in cm per second) at that instant is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

किसी वृत्त के एक त्रिज्यखण्ड का क्षेत्रफल अचर $25\ \text{cm}^2$ है, जबकि इसकी त्रिज्या r तथा केन्द्रीय कोण समय के साथ परिवर्तित होते हैं। जिस क्षण $r = 4$ cm है, उस क्षण केन्द्रीय कोण $\frac{45}{\pi}$ अंश प्रति सेकण्ड की दर से घट रहा है। तब उस क्षण त्रिज्यखण्ड के परिमाप के परिवर्तन की दर (cm प्रति सेकण्ड में) _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Work with the central angle θ in radians, since the formulas $s = r\theta$ (arc) and $\frac{1}{2}r^2\theta$ (area) hold only in radians. $1^\circ = \frac{\pi}{180}$ rad, so the given rate is $\frac{d\theta}{dt} = -\frac{45}{\pi} \cdot \frac{\pi}{180} = -\frac{1}{4}$ rad per second.

Constant area: $\frac{1}{2}r^2\theta = 25 \Rightarrow \theta = \frac{50}{r^2}$. At $r = 4$: $\theta = \frac{50}{16} = \frac{25}{8}$ radian, which is less than 2π, so the sector exists.

Differentiating $\theta = 50r^{-2}$: $\frac{d\theta}{dt} = -\frac{100}{r^3}\frac{dr}{dt}$. At $r = 4$: $-\frac{1}{4} = -\frac{100}{64}\frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{64}{400} = 0.16$ cm/s (the radius grows as the angle shrinks).

Perimeter = two radii + arc: $P = 2r + r\theta = 2r + \frac{50}{r}$.

$$\frac{dP}{dt} = \left(2 - \frac{50}{r^2}\right)\frac{dr}{dt} = \left(2 - \frac{50}{16}\right)(0.16) = \left(-\frac{9}{8}\right)(0.16) = -0.18 \text{ cm/s}.$$

So the perimeter is decreasing at 0.18 cm/s at that instant.

Final Answer: -0.18

Question 14

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \dfrac{x}{1+x^2}$. For each $t \in \mathbb{R}$, let $g(t)$ denote the least value of f on the closed interval $[t, t+1]$. If M is the greatest value of $g(t)$ as t varies over $\mathbb{R}$, then the value of M^2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \dfrac{x}{1+x^2}$ द्वारा परिभाषित है। प्रत्येक $t \in \mathbb{R}$ के लिए, माना $g(t)$ संवृत अन्तराल $[t, t+1]$ पर f के न्यूनतम मान को दर्शाता है। यदि t के $\mathbb{R}$ पर परिवर्तित होने पर $g(t)$ का अधिकतम मान M है, तो M^2 का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$f'(x) = \dfrac{(1+x^2) - 2x^2}{(1+x^2)^2} = \dfrac{1-x^2}{(1+x^2)^2}$. So f is decreasing on $(-\infty, -1]$, increasing on $[-1, 1]$ and decreasing on $[1, \infty)$.

$f''(x) = \dfrac{2x(x^2-3)}{(1+x^2)^3}$, so $f''(1) = -\dfrac{1}{2} < 0$ and $f''(-1) = \dfrac{1}{2} > 0$: local maximum $f(1) = \dfrac{1}{2}$ and local minimum $f(-1) = -\dfrac{1}{2}$. Also $f(x) > 0$ for $x > 0$ and $f(x) \le 0$ for $x \le 0$.

$g(t)$ is the absolute minimum of the continuous f on the closed interval $[t, t+1]$, attained at an endpoint or at a critical point inside.

Case $t \le 0$: $t \in [t, t+1]$ and $f(t) \le 0$, so $g(t) \le 0$.

Case $t \ge 1$: f is decreasing on $[t, t+1]$, so $g(t) = f(t+1)$, which is decreasing in t; hence $g(t) \le f(2) = \dfrac{2}{5}$.

Case $0 < t < 1$: the only critical point inside $[t, t+1]$ is $x = 1$, a maximum, so the minimum is at an endpoint: $g(t) = \min\{f(t), f(t+1)\}$. On $(0,1)$, $f(t)$ is increasing in t and $f(t+1)$ is decreasing in t, so g increases while $f(t) < f(t+1)$ and decreases afterwards; its greatest value is at $f(t) = f(t+1)$.

$\dfrac{t}{1+t^2} = \dfrac{t+1}{t^2+2t+2} \Rightarrow t^3 + 2t^2 + 2t = t^3 + t^2 + t + 1 \Rightarrow t^2 + t - 1 = 0$, so $t = \dfrac{\sqrt{5}-1}{2} \in (0,1)$.

Using $t^2 = 1 - t$: $1 + t^2 = 2 - t$ and $f(t) = \dfrac{t}{2-t} = \dfrac{\sqrt{5}-1}{5-\sqrt{5}} = \dfrac{1}{\sqrt{5}} \approx 0.447$, which exceeds $\dfrac{2}{5}$ from the case $t \ge 1$ and 0 from $t \le 0$.

Therefore $M = \dfrac{1}{\sqrt{5}}$ and $M^2 = \dfrac{1}{5} = 0.2$.

Final Answer: 0.2

Question 15

English

Let S be the set of all ordered pairs (a, b), where $a, b \in \{-10, -9, \ldots, 9, 10\}$, for which the function $f : [0, \infty) \to \mathbb{R}$ defined by $f(x) = x^3 + ax^2 + bx$ is one-one. Then the number of elements in S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S उन सभी क्रमित युग्मों (a, b) का समुच्चय है, जहाँ $a, b \in \{-10, -9, \ldots, 9, 10\}$, जिनके लिए $f(x) = x^3 + ax^2 + bx$ द्वारा परिभाषित फलन $f : [0, \infty) \to \mathbb{R}$ एकैकी है। तब S में अवयवों की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A continuous one-one function on an interval is strictly monotonic. Since $f(x) \to \infty$ as $x \to \infty$, a one-one f on $[0, \infty)$ must be strictly increasing. Conversely, if $f' \geq 0$ on $[0, \infty)$ with f' vanishing only at isolated points, f is strictly increasing and hence one-one.

So the condition is $f'(x) = 3x^2 + 2ax + b \geq 0$ for all $x \geq 0$ (a quadratic that is not identically zero has isolated zeros). If $f'(x_1) < 0$ for some $x_1 \geq 0$, f decreases near x_1 and later increases, so it takes some value twice and is not one-one.

The minimum of f' on $[0, \infty)$: the vertex is at $x = -\dfrac{a}{3}$.

Case $a \geq 0$: the vertex is at $x \leq 0$, so f' is increasing on $[0, \infty)$ and the condition is $f'(0) = b \geq 0$. With $a \in \{0, \ldots, 10\}$ and $b \in \{0, \ldots, 10\}$: 121 pairs.

Case $a < 0$: the vertex $-\dfrac{a}{3} > 0$ and the minimum value is $b - \dfrac{a^2}{3}$, so the condition is $3b \geq a^2$ (equality gives $f' = 3\left(x + \frac{a}{3}\right)^2$, still allowed).

$a = -1$: $b \geq \frac{1}{3}$, $b \in \{1, \ldots, 10\}$: 10. $a = -2$: $b \geq \frac{4}{3}$: 9. $a = -3$: $b \geq 3$: 8. $a = -4$: $b \geq \frac{16}{3}$, $b \in \{6, \ldots, 10\}$: 5. $a = -5$: $b \geq \frac{25}{3}$, $b \in \{9, 10\}$: 2. $a \leq -6$: $b \geq 12$, none. Total 34.

Number of elements of S: $121 + 34 = 155$.

Final Answer: 155

Question 16

English

Let α be the least real number and β the greatest real number such that

$$\frac{x}{(1+x)^\alpha} \leq \log_e(1+x) \leq \frac{x}{(1+x)^\beta}$$

for all $x \geq 0$. Then the value of $\alpha + \beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि α वह न्यूनतम वास्तविक संख्या तथा β वह अधिकतम वास्तविक संख्या है जिनके लिए

$$\frac{x}{(1+x)^\alpha} \leq \log_e(1+x) \leq \frac{x}{(1+x)^\beta}$$

सभी $x \geq 0$ के लिए सत्य है। तब $\alpha + \beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Series for $|x| < 1$: $\log_e(1+x) = x - \dfrac{x^2}{2} + \dfrac{x^3}{3} - \cdots$, and by the binomial series $\dfrac{x}{(1+x)^k} = x\left(1 - kx + \dfrac{k(k+1)}{2}x^2 - \cdots\right) = x - kx^2 + O(x^3)$.

Upper bound. If $\beta > \dfrac{1}{2}$, then $\dfrac{x}{(1+x)^\beta} - \log_e(1+x) = \left(\dfrac{1}{2} - \beta\right)x^2 + O(x^3) < 0$ for small $x > 0$, so the inequality fails. Hence $\beta \le \dfrac{1}{2}$.

For $\beta = \dfrac{1}{2}$, let $h(x) = \dfrac{x}{\sqrt{1+x}} - \log_e(1+x)$ on $[0, \infty)$. Then $h'(x) = \dfrac{1}{\sqrt{1+x}} - \dfrac{x}{2(1+x)^{3/2}} - \dfrac{1}{1+x} = \dfrac{x + 2 - 2\sqrt{1+x}}{2(1+x)^{3/2}} = \dfrac{\left(\sqrt{1+x} - 1\right)^2}{2(1+x)^{3/2}} \ge 0$.

So h is increasing and $h(x) \ge h(0) = 0$: $\log_e(1+x) \le \dfrac{x}{\sqrt{1+x}}$ for all $x \ge 0$. For $\beta < \dfrac{1}{2}$ the right side is even larger. So the greatest β is $\dfrac{1}{2}$.

Lower bound. The series only gives $\alpha \ge \dfrac{1}{2}$ near 0; the behaviour for large x decides. If $0 \le \alpha < 1$, then for $x \ge 1$, $(1+x)^\alpha \le (2x)^\alpha \le 2x^\alpha$, so $\dfrac{x}{(1+x)^\alpha} \ge \dfrac{x^{1-\alpha}}{2}$ (and if $\alpha < 0$, directly $\dfrac{x}{(1+x)^\alpha} \ge x^{1-\alpha}$), while $\dfrac{\log_e(1+x)}{x^{1-\alpha}} \to 0$ as $x \to \infty$. So the left inequality fails for large x, and $\alpha \ge 1$.

For $\alpha = 1$, let $u(x) = \log_e(1+x) - \dfrac{x}{1+x}$. Then $u'(x) = \dfrac{1}{1+x} - \dfrac{1}{(1+x)^2} = \dfrac{x}{(1+x)^2} \ge 0$ and $u(0) = 0$, so $u \ge 0$ on $[0, \infty)$. For $\alpha > 1$ the left side is smaller still. So the least α is 1.

Hence $\alpha + \beta = 1 + \dfrac{1}{2} = 1.5$.

Final Answer: 1.5

Question 17

English

Let k be a real number and let $f : (0, \infty) \to \mathbb{R}$ be defined by $f(x) = x + \frac{k}{x}$. Suppose f satisfies all the conditions of Rolle's theorem on the interval $[1, 4]$. If c is the point of $\left(\frac{1}{4}, 25\right)$ given by Lagrange's mean value theorem for f on $\left[\frac{1}{4}, 25\right]$, then the value of c is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि k एक वास्तविक संख्या है तथा $f : (0, \infty) \to \mathbb{R}$, $f(x) = x + \frac{k}{x}$ द्वारा परिभाषित है। मान लीजिए कि f अन्तराल $[1, 4]$ पर रोले प्रमेय की सभी शर्तों को सन्तुष्ट करता है। यदि f के लिए अन्तराल $\left[\frac{1}{4}, 25\right]$ पर लग्रांज माध्यमान प्रमेय द्वारा प्राप्त $\left(\frac{1}{4}, 25\right)$ का बिन्दु c है, तो c का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

On $[1, 4] \subset (0, \infty)$, f is continuous and differentiable for every k, so the only Rolle condition that restricts k is $f(1) = f(4)$: $1 + k = 4 + \frac{k}{4} \Rightarrow \frac{3k}{4} = 3 \Rightarrow k = 4$. (Then $f'(x) = 1 - \frac{4}{x^2} = 0$ at $x = 2 \in (1, 4)$.)

So $f(x) = x + \frac{4}{x}$, which is continuous on $\left[\frac{1}{4}, 25\right]$ and differentiable on $\left(\frac{1}{4}, 25\right)$, so Lagrange's mean value theorem applies.

For $0 < a < b$: $\frac{f(b) - f(a)}{b - a} = \frac{(b - a) + 4\left(\frac{1}{b} - \frac{1}{a}\right)}{b - a} = 1 - \frac{4}{ab}$.

With $a = \frac{1}{4}$, $b = 25$: the chord slope is $1 - \frac{4}{25/4} = 1 - \frac{16}{25} = \frac{9}{25}$ (directly: $\frac{25.16 - 16.25}{24.75} = 0.36$).

$f'(c) = 1 - \frac{4}{c^2} = \frac{9}{25} \Rightarrow c^2 = \frac{25}{4} \Rightarrow c = \frac{5}{2}$, since c must lie in $\left(\frac{1}{4}, 25\right)$; c is the geometric mean of the end points.

Final Answer: 2.5

Question 18

English

Let $P(x) = (x+2)(x+1)(2x-1)(x-3)$, $x \in \mathbb{R}$. Then the sum of all the real roots of the equation $P'(x) + 2P(x) = 0$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $P(x) = (x+2)(x+1)(2x-1)(x-3)$, $x \in \mathbb{R}$। तब समीकरण $P'(x) + 2P(x) = 0$ के सभी वास्तविक मूलों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$P(x) = (x+2)(x+1)(2x-1)(x-3) = (x^2+3x+2)(2x^2-7x+3) = 2x^4 - x^3 - 14x^2 - 5x + 6$, so $P'(x) = 8x^3 - 3x^2 - 28x - 5$.

$Q(x) = P'(x) + 2P(x) = 4x^4 + 6x^3 - 31x^2 - 38x + 7$, a quartic with leading coefficient 4.

Let $h(x) = e^{2x}P(x)$. Then $h'(x) = e^{2x}\left(P'(x) + 2P(x)\right) = e^{2x}Q(x)$, and $e^{2x} > 0$, so the roots of Q are exactly the roots of h'.

h is differentiable and $h(-2) = h(-1) = h\left(\frac{1}{2}\right) = h(3) = 0$. Rolle's theorem on $[-2,-1]$, $\left[-1,\frac{1}{2}\right]$ and $\left[\frac{1}{2},3\right]$ gives roots of Q in $(-2,-1)$, $\left(-1,\frac{1}{2}\right)$ and $\left(\frac{1}{2},3\right)$: three distinct real roots.

Fourth root: $Q(-2) = P'(-2) + 2P(-2) = P'(-2)$. Since $P'(-2) = (-2+1)(2(-2)-1)(-2-3) = (-1)(-5)(-5) = -25$ (the other terms of the product rule vanish at the root -2), $Q(-2) = -25 < 0$. Also $Q(x) \to +\infty$ as $x \to -\infty$. By the intermediate value theorem Q has a root in $(-\infty,-2)$.

So Q has four distinct real roots, one in each of $(-\infty,-2)$, $(-2,-1)$, $\left(-1,\frac{1}{2}\right)$, $\left(\frac{1}{2},3\right)$, and these are all its roots.

The sum of all real roots is therefore the sum of all roots: $-\dfrac{6}{4} = -1.5$ (numerically the roots are about -2.966, -1.358, 0.163, 2.661).

Final Answer: -1.5

Question 19

English

Let S be the set of all ordered pairs (a, b) of integers for which the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = ax + b\sin x$ is one-one and satisfies $|f(x) - f(y)| \leq 6|x - y|$ for all $x, y \in \mathbb{R}$. Then the number of elements in S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S पूर्णांकों के उन सभी क्रमित युग्मों (a, b) का समुच्चय है जिनके लिए $f(x) = ax + b\sin x$ द्वारा परिभाषित फलन $f : \mathbb{R} \to \mathbb{R}$ एकैकी है तथा सभी $x, y \in \mathbb{R}$ के लिए $|f(x) - f(y)| \leq 6|x - y|$ को सन्तुष्ट करता है। तब S में अवयवों की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$f'(x) = a + b\cos x$. As $\cos x$ runs over $[-1, 1]$, f' takes every value in $[a - |b|, a + |b|]$, so $\max|f'| = |a| + |b|$.

Lipschitz condition. If $|f'(t)| \leq 6$ for all t, then by Lagrange's mean value theorem $|f(x) - f(y)| = |f'(\xi)||x - y| \leq 6|x - y|$. Conversely, if $|f(x) - f(y)| \leq 6|x - y|$ for all x, y, then $\left|\dfrac{f(x) - f(y)}{x - y}\right| \leq 6$, and letting $y \to x$ gives $|f'(x)| \leq 6$. So the condition is $|a| + |b| \leq 6$.

One-one condition. A continuous one-one function on $\mathbb{R}$ is strictly monotonic. If $|b| > |a|$, then $a - |b| < 0 < a + |b|$, so f' takes both signs on intervals and f is not monotonic, hence many-one. If $a = b = 0$, f is constant.

If $|a| \geq |b|$ and $a \neq 0$, then $f'(x) = a + b\cos x$ has the sign of a except at isolated points where it may vanish (when $|a| = |b|$), so f is strictly monotonic and one-one (by the mean value theorem $f(x) - f(y) = f'(\xi)(x - y) \neq 0$ for $x \neq y$, since f' cannot vanish on a whole interval).

So $S = \{(a, b) \in \mathbb{Z}^2 : a \neq 0,\ |b| \leq |a|,\ |a| + |b| \leq 6\}$.

For $|a| = k$: $|b| \leq \min(k, 6 - k)$, giving $2\min(k, 6 - k) + 1$ values: $k = 1 : 3$, $k = 2 : 5$, $k = 3 : 7$, $k = 4 : 5$, $k = 5 : 3$, $k = 6 : 1$, total 24.

Each k has two values of a ($\pm k$), so $|S| = 2 \times 24 = 48$.

Final Answer: 48

Question 20

English

Let $g : \mathbb{R} \to \mathbb{R}$ and $h : \mathbb{R} \to \mathbb{R}$ be defined by $g(x) = \dfrac{8x}{1+x^2}$ and $h(x) = 2\left(g(x)\right)^3 - 3\left(g(x)\right)^2 - 36g(x)$. Then the number of integers in the range of h is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $g : \mathbb{R} \to \mathbb{R}$ तथा $h : \mathbb{R} \to \mathbb{R}$, $g(x) = \dfrac{8x}{1+x^2}$ तथा $h(x) = 2\left(g(x)\right)^3 - 3\left(g(x)\right)^2 - 36g(x)$ द्वारा परिभाषित हैं। तब h के परिसर में पूर्णांकों की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Bound on g: $1 + x^2 - 2|x| = (|x| - 1)^2 \geq 0$, so $\left|\dfrac{8x}{1+x^2}\right| \leq \dfrac{8|x|}{2|x|} = 4$ for $x \neq 0$ (and $g(0) = 0$). Equality holds at $x = \pm 1$: $g(1) = 4$, $g(-1) = -4$. So the greatest and least values of g are 4 and -4.

g is continuous on $\mathbb{R}$, so by the intermediate value property its range is the whole interval $[-4, 4]$.

Write $h(x) = \varphi(g(x))$ with $\varphi(t) = 2t^3 - 3t^2 - 36t$. Then the range of h is $\varphi([-4, 4])$.

$\varphi'(t) = 6t^2 - 6t - 36 = 6(t-3)(t+2)$, so the critical points in $[-4, 4]$ are $t = -2$ and $t = 3$.

Values: $\varphi(-4) = -128 - 48 + 144 = -32$, $\varphi(-2) = -16 - 12 + 72 = 44$, $\varphi(3) = 54 - 27 - 108 = -81$, $\varphi(4) = 128 - 48 - 144 = -64$.

So on the closed interval $[-4, 4]$, the absolute maximum of φ is 44 (at $t = -2$) and the absolute minimum is -81 (at $t = 3$). Since φ is continuous, $\varphi([-4, 4]) = [-81, 44]$, which is the range of h.

The integers in $[-81, 44]$ number $44 - (-81) + 1 = 126$.

Final Answer: 126

Question 21

English

Let S be the sphere $x^2 + y^2 + z^2 = 25$ and let Q be the point $(6, 7, 6)$. For a real number d with $|d| < 55$, the plane $6x + 7y + 6z = d$ cuts S in a circle C_d. Among all right circular cones with vertex Q and base C_d, the cone of maximum volume is obtained for d equal to _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S गोला $x^2 + y^2 + z^2 = 25$ है तथा Q बिन्दु $(6, 7, 6)$ है। किसी वास्तविक संख्या d, $|d| < 55$, के लिए तल $6x + 7y + 6z = d$ गोले S को एक वृत्त C_d में काटता है। शीर्ष Q तथा आधार C_d वाले सभी लम्ब वृत्तीय शंकुओं में से अधिकतम आयतन वाला शंकु d के _____ के बराबर होने पर प्राप्त होता है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The centre of S is $O(0, 0, 0)$ and its radius is 5. $|\overrightarrow{OQ}| = \sqrt{6^2 + 7^2 + 6^2} = 11 > 5$, so Q is outside S.
The normal of the plane $6x + 7y + 6z = d$ is $(6, 7, 6)$, parallel to $\overrightarrow{OQ}$. The centre of C_d is the foot of the perpendicular from O, $\dfrac{d}{121}(6, 7, 6)$, at signed distance $u = \dfrac{d}{11}$ from O along $\overrightarrow{OQ}$. So O, the centre of C_d and Q are collinear on the axis of C_d, and every such cone is right circular.
The plane cuts S iff $|u| < 5$, i.e. $|d| < 55$. Radius of C_d: $r^2 = 25 - u^2$. Height of the cone (distance from Q to the plane): $11 - u$.
Volume: $V(u) = \dfrac{\pi}{3}(25 - u^2)(11 - u) = \dfrac{\pi}{3}\left(u^3 - 11u^2 - 25u + 275\right)$, $u \in (-5, 5)$.
$V'(u) = \dfrac{\pi}{3}\left(3u^2 - 22u - 25\right) = \dfrac{\pi}{3}(3u - 25)(u + 1)$. The root $u = \dfrac{25}{3}$ is outside $(-5, 5)$, so the only critical point is $u = -1$.
$V' > 0$ on $(-5, -1)$ and $V' < 0$ on $(-1, 5)$, so $u = -1$ gives the maximum volume $V = \dfrac{\pi}{3}\cdot 24\cdot 12 = 96\pi$ (the base lies on the far side of the centre from Q).
Hence $d = 11u = -11$.

Final Answer: -11

Question 22

English

Let $F : \mathbb{R} \to \mathbb{R}$ be defined by $F(x) = 4x^4 - 2x^3 - 3x^2$, and let $g : \mathbb{R} \to \mathbb{R}$ be defined by $g(x) = \max\{F(x), F(-x)\}$. Then the distance between the points of inflection of the curve $y = g(x)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $F : \mathbb{R} \to \mathbb{R}$, $F(x) = 4x^4 - 2x^3 - 3x^2$ द्वारा परिभाषित है, तथा $g : \mathbb{R} \to \mathbb{R}$, $g(x) = \max\{F(x), F(-x)\}$ द्वारा परिभाषित है। तब वक्र $y = g(x)$ के नति-परिवर्तन बिन्दुओं (points of inflection) के बीच की दूरी _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$F(x) - F(-x) = (4x^4 - 2x^3 - 3x^2) - (4x^4 + 2x^3 - 3x^2) = -4x^3$.

For $x > 0$ this is negative, so $g(x) = F(-x)$; for $x < 0$ it is positive, so $g(x) = F(x) = F(-|x|)$. Hence $g(x) = F(-|x|) = 4x^4 + 2|x|^3 - 3x^2$ for all x: the graph is the part of $y = F(x)$ for $x \le 0$ together with its mirror image in the y-axis (it is $F(-|x|)$, not $F(|x|)$).

g is even. For $x > 0$, $g'(x) = 16x^3 + 6x^2 - 6x$ and $g''(x) = 48x^2 + 12x - 6$; by symmetry $g''(x) = 48x^2 + 12|x| - 6$ for all x, and g is twice differentiable at 0 ($g'(0) = 0$, $g''(0) = -6$ from both sides).

$48u^2 + 12u - 6 = 6(8u^2 + 2u - 1) = 6(4u - 1)(2u + 1)$ with $u = |x|$, so $g''(x) < 0$ for $|x| < \frac{1}{4}$ and $g''(x) > 0$ for $|x| > \frac{1}{4}$.

Thus the concavity changes exactly at $x = \pm\frac{1}{4}$; at $x = 0$ the curve is concave down on both sides (a local maximum), so it is not a point of inflection. (Using $F(|x|)$ by mistake would give $x = \pm\frac{1}{2}$ and the wrong distance 1.)

The inflection points are $\left(\pm\frac{1}{4}, g\left(\frac{1}{4}\right)\right)$ with $g\left(\frac{1}{4}\right) = \frac{4}{256} + \frac{2}{64} - \frac{3}{16} = -\frac{9}{64}$; they have equal ordinates, so the distance between them is $\frac{1}{4} - \left(-\frac{1}{4}\right) = \frac{1}{2} = 0.5$.

Final Answer: 0.5